

12.1 Cyclone Debbie impact ranking

Cyclone Debbie caused a range of hydrological and atmospheric impacts across Queensland and northern NSW. These impacts affected people, environments and economies in different ways. Your task is to read the cards that describe the major impacts of Cyclone Debbie. You will then use **specific criteria** (such as severity or extent) to **judge and rank** which impacts were the most significant.

Criteria 1: Number of people affected

Rank	Impact	Justification
1 (most severe)		
2		
3		
4		
5		
6 (least severe)		

Criteria 2: Severity of damage

Rank	Impact	Justification
1 (most severe)		
2		
3		
4		
5		
6 (least severe)		

Criteria 3: Long-term consequences

Rank	Impact	Justification
1 (most severe)		
2		
3		
4		
5		
6 (least severe)		

Support: The following sentence starters can be used to support your rankings

- “The _____ impact is ranked highest because...”
- “Although _____ caused immediate damage, _____ had longer-term consequences...”
- “This impact affected a larger spatial area than...”

Extension: Explain which impact would be hardest to manage in the future and why.

(cards are found on the following page)

Impact Card 1 — Extreme rainfall and river flooding During Cyclone Debbie, extremely heavy rainfall fell across large parts of Queensland and northern New South Wales. Some locations received more than 1,000 mm of rain over several days, which overwhelmed river catchments and drainage systems. As rivers rapidly filled and overflowed, widespread flooding occurred in both urban and rural areas. Homes, roads and businesses were inundated, and many communities were cut off for days or weeks. The flooding continued well after the cyclone itself weakened, meaning recovery was slow and costly. This impact was especially severe because it affected a large geographic area and caused long-term disruption to people's lives.	Impact Card 2 — Storm surge and coastal flooding As Cyclone Debbie approached the coast near Airlie Beach, strong onshore winds and very low air pressure pushed large volumes of seawater towards the land. This created a storm surge, which caused coastal flooding along parts of the Whitsunday coastline. Seawater flooded low-lying coastal areas, damaged roads and buildings, and eroded beaches. In some places, saltwater entered freshwater systems and coastal ecosystems, causing environmental damage. Because sea levels are higher today than in the past, the storm surge travelled further inland, increasing the extent and severity of coastal flooding.	Impact Card 3 — Wind damage and infrastructure failure Cyclone Debbie made landfall as a powerful Category 4 system, producing extremely strong winds with gusts exceeding 260 km/h in some areas. These winds damaged or destroyed thousands of homes and caused widespread damage to infrastructure such as power lines, communication towers and transport networks. Large areas lost electricity and phone services, sometimes for weeks. The loss of infrastructure made it difficult for emergency services to respond quickly and slowed recovery efforts. Wind damage was one of the most immediate and visible impacts, posing serious risks to safety and leaving communities vulnerable in the days following the cyclone.
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Impact Card 4 — Landslides and environmental damage The intense rainfall associated with Cyclone Debbie caused soils in steep and hilly areas to become saturated, triggering landslides. These occurred in parts of Queensland and northern New South Wales, particularly in hinterland and mountainous regions. Landslides blocked roads, damaged vegetation and isolated communities, making it harder for emergency services and supplies to reach affected areas. Even after rainfall ended, slopes remained unstable, creating ongoing safety risks. This impact added to environmental damage and increased the cost and complexity of recovery.	Impact Card 5 — Agricultural damage and economic loss Cyclone Debbie caused severe damage to agricultural regions, particularly in Queensland's sugar-growing areas. Strong winds flattened crops, while flooding destroyed farmland and drowned livestock. More than 70 per cent of Queensland's sugar crop was damaged, resulting in significant financial losses for farmers. Many agricultural areas remained waterlogged for long periods, delaying replanting and reducing future yields. These losses affected not only individual farmers but also food supply chains, employment and export income, extending the economic impact of the cyclone well beyond the immediate disaster period.	Impact Card 6 — Social and economic disruption The combined effects of flooding, wind damage, infrastructure failure and agricultural loss caused widespread social and economic disruption. Schools, hospitals and businesses were forced to close, and tourism in regions such as the Whitsundays declined sharply. The total cost of damage from Cyclone Debbie exceeded \$3.5 billion, placing pressure on governments, insurance systems and communities. Many people experienced long-term stress, displacement and financial hardship during recovery. This impact highlights how cyclones can affect not just physical environments, but also wellbeing and economic stability over time.
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